University of Asia Pacific

Department of Computer Science & Engineering

Course Name: Computer Networks Lab

Course Code: CSE 320

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Introduction

The **Vic Modern Hotel** requires a **secure, scalable, and efficient** network infrastructure to support its three-floor operations, including departments like Reception, Finance, HR, IT, and Admin. This project designs and simulates the hotel's network using **Cisco Packet Tracer**, ensuring:

- **Segmented VLANs** for departmental isolation.
- **OSPF dynamic routing** for inter-floor communication.
- **Wi-Fi access** for guests and staff.
- **DHCP & SSH** for automated IP assignment and secure remote management.

Objective:

- Build a fault-tolerant network where all devices communicate seamlessly.
- Implement security measures (port security, VLANs, SSH).

Motivation

Modern hotels rely on robust networks for:

- **Guest services** (Wi-Fi, digital check-in).
- **Staff operations** (POS, VoIP, surveillance).
- **Security** (VLAN segregation, port security).

Why Cisco Packet Tracer?

- Simulates enterprise networks without physical hardware.
- Validates configurations (OSPF, DHCP, VLANs) before real-world deployment.

Methodology

Network Topology

- **3 Routers** (1 per floor) connected via **Serial DCE** (10.10.10.0/30 subnets).
- **3 Switches** (1 per floor) with **VLANs**:
  - **1st Floor**: Reception (VLAN 80), Store (VLAN 70), Logistics (VLAN 60).
  - **2nd Floor**: Finance (VLAN 50), HR (VLAN 40), Sales (VLAN 30).
  - **3rd Floor**: IT (VLAN 10), Admin (VLAN 20).
- **Wi-Fi**: Separate SSIDs for staff/guests.
- **Printers**: 1 per department.

Configurations

- **Turn on Routers:**

```
Router(config)#int se0/2/0
Router(config-if)#no sh

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
Router(config-if)#int se0/2/1
Router(config-if)#no sh

%LINK-5-CHANGED: Interface Serial0/2/1, changed state to down

Router(config-if)#int gig0/0
Router(config-if)#no sh
```

- **Enable Clock Rate into Serial DCE Interphases:**

Enabling clock rate to some particular (clock signed) interphases is required

```
Router(config-if)#int se0/0/1
Router(config-if)#clock rate 64000
Router(config-if)#
```

- **VLANs**:

(1st Floor)

```
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#int range fa0/2-3
Switch(config-if-range)#switc
Switch(config-if-range)#switchport mo
Switch(config-if-range)#switchport mode ac
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#swit
Switch(config-if-range)#switchport acc
Switch(config-if-range)#switchport access vlan 80
% Access VLAN does not exist. Creating vlan 80
Switch(config-if-range)#
Switch(config-if-range)#
Switch(config-if-range)#
Switch(config-if-range)#int range fa0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 70
% Access VLAN does not exist. Creating vlan 70
Switch(config-if-range)#
Switch(config-if-range)#int range fa0/6-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 60
% Access VLAN does not exist. Creating vlan 60
Switch(config-if-range)#
Switch(config-if-range)#
Switch(config-if-range)#do wr
Building configuration...
[OK]
Switch(config-if-range)#
```

(2nd Floor)

```
Switch>
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int range fa0/2-3
Switch(config-if-range)#swit
Switch(config-if-range)#switchport mo
Switch(config-if-range)#switchport mode ac
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#swit
Switch(config-if-range)#switchport acc
Switch(config-if-range)#switchport access vlan 50
% Access VLAN does not exist. Creating vlan 50
Switch(config-if-range)#
Switch(config-if-range)#
Switch(config-if-range)#int range fa0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
% Access VLAN does not exist. Creating vlan 40
Switch(config-if-range)#
Switch(config-if-range)#
Switch(config-if-range)#
Switch(config-if-range)#int range fa0/6-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
% Access VLAN does not exist. Creating vlan 30
Switch(config-if-range)#do wr
Building configuration...
[OK]
Switch(config-if-range)#
```

(3rd Floor)

```
Switch(config)#int range fa0/2-3
Switch(config-if-range)#swit
Switch(config-if-range)#switchport mo
Switch(config-if-range)#switchport mode ac
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switc
Switch(config-if-range)#switchport acc
Switch(config-if-range)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10
Switch(config-if-range)#
Switch(config-if-range)#
Switch(config-if-range)#
Switch(config-if-range)#int range fa0/4-6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
```

Trunking:

```
Switch(config-if-range)#int range fa0/1
Switch(config-if-range)#swit
Switch(config-if-range)#switchport mo
Switch(config-if-range)#switchport mode tr
Switch(config-if-range)#switchport mode trunk
```

- **Inter-Router IP Configuration:**

```
Router(config-if)#in se0/2/0
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#ip add
Router(config-if)#ip address 10.10.10.5 255.255.255.252
Router(config-if)#
Router(config-if)#
Router(config-if)#in se0/2/1
Router(config-if)#ip address 10.10.10.9 255.255.255.252
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#int se0/1/1
Router(config-if)#clo
Router(config-if)#clock ra
Router(config-if)#clock rate 64000
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#int se0/1/0
Router(config-if)#ip add
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#int se0/1/1
Router(config-if)#ip address 10.10.10.10 255.255.255.252
Router(config-if)#do wr
Building configuration...
[OK]
Router(config)#int se0/2/0
Router(config-if)#ip add
Router(config-if)#ip address 10.10.10.6 255.255.255.252
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#int se0/2/1
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#do wr
Building configuration...
[OK]
```

- **Inter-VLAN Configuration:**

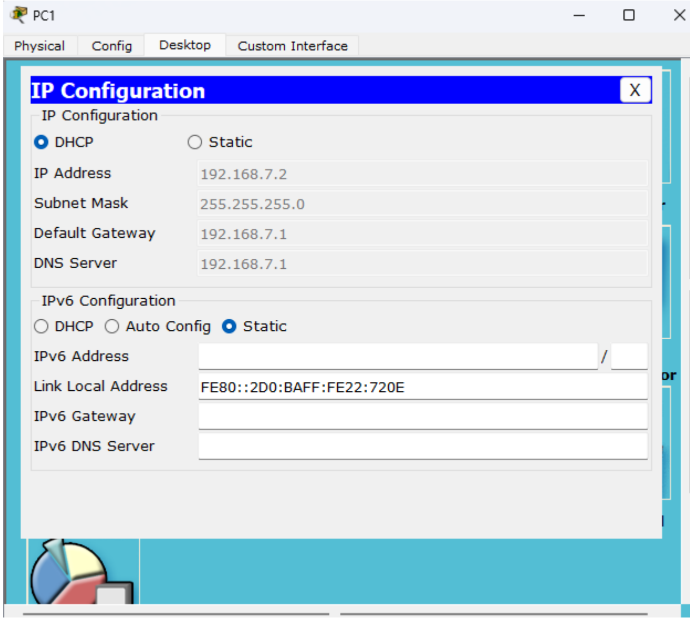
```
Router(config)#int fa0/0.80
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.80, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.80, changed state
to up

Router(config-subif)#encapsulation dot1Q 80
Router(config-subif)#ip address 192.168.8.1 255.255.255.0
Router(config-subif)#ex
Router(config)#int fa0/0.70
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.70, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.70, changed state
to up

Router(config-subif)#encapsulation dot1Q 70
Router(config-subif)#ip address 192.168.7.1 255.255.255.0
Router(config-subif)#ex
Router(config)#
Router(config)#
Router(config)#int fa0/0.60
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.60, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.60, changed state
to up

Router(config-subif)#encapsulation dot1Q 60
Router(config-subif)#ip address 192.168.6.1 255.255.255.0
Router(config-subif)#do wr
Building configuration...
[OK]
```

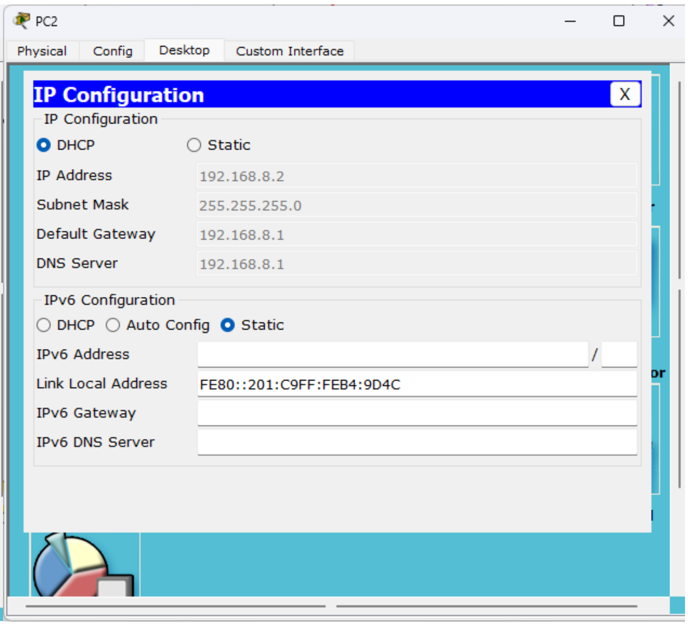
(Only for the router of the 1st floor are being shown here)

- **DHCP**:

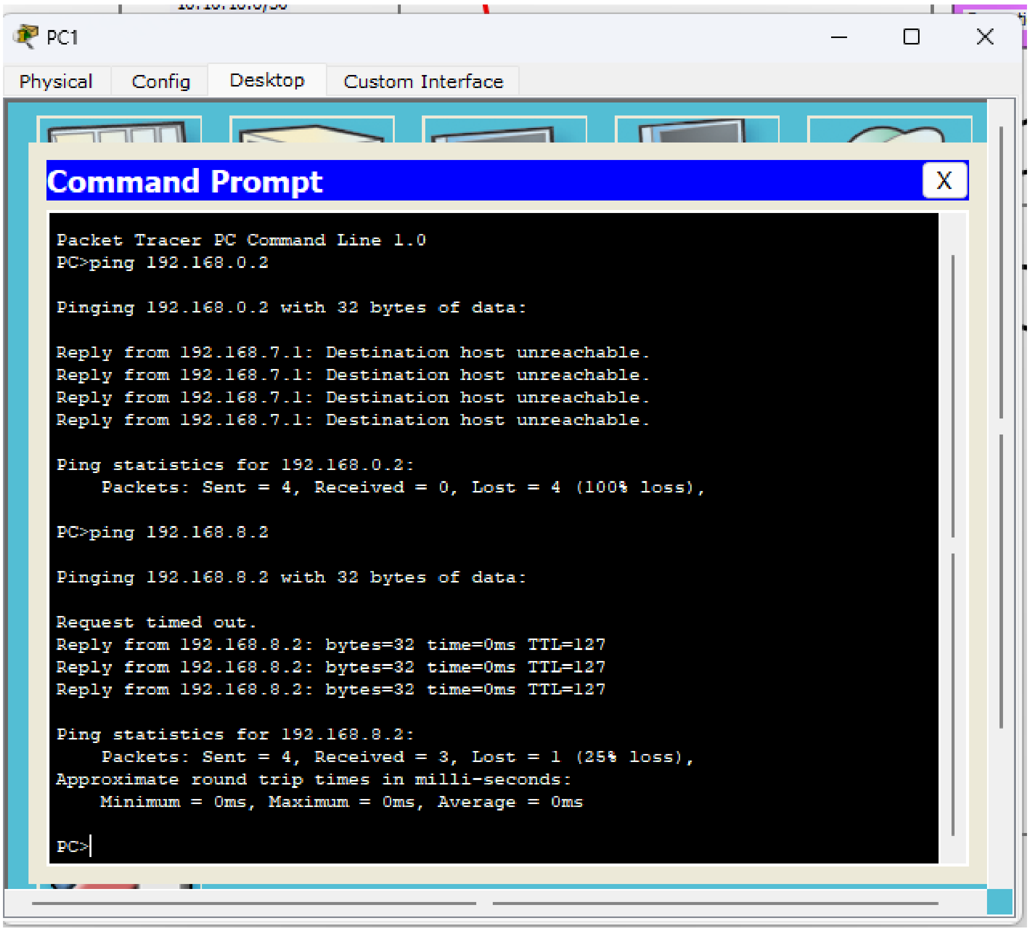
```
Router(config)#service dhcp
Router(config)#ip dhcp pool Reception
Router(dhcp-config)#network 192.168.8.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.8.1
Router(dhcp-config)#dns-server 192.168.8.1
Router(dhcp-config)#ex
Router(config)#
Router(config)#ip dhcp pool Store
Router(dhcp-config)#network 192.168.7.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.7.1
Router(dhcp-config)#dns-server 192.168.7.1
Router(dhcp-config)#ex
Router(config)#
Router(config)#
Router(config)#ip dhcp pool Logistics
Router(dhcp-config)#network 192.168.6.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.6.1
Router(dhcp-config)#dns-server 192.168.6.1
Router(dhcp-config)#ex
Router(config)#do wr
Building configuration...
[OK]
```

(Only for the router of the 1st floor are being shown here)





As a result, after enabling DHCP, the ip address for PCs and Printers will being automatically generated via the system.



Transaction between intra-floor VLANs PCs were connectable but unreachable for inter-floor connections.

- **OSPF Routing**:

(1st Floor)

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 10
Router(config-router)#network 10.10.10.4 255.255.255.252 area 0
Router(config-router)#network 10.10.10.8 255.255.255.252 area 0
Router(config-router)#network 192.168.8.0 255.255.255.0 area 0
Router(config-router)#network 192.168.7.0 255.255.255.0 area 0
Router(config-router)#network 192.168.6.0 255.255.255.0 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
00:22:16: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.5.1 on Serial0/0/1 from LOADING
to FULL, Loading Done

00:24:25: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.2.1 on Serial0/0/0 from LOADING
to FULL, Loading Done
```

(2nd Floor)

```
Router(config)#router ospf 10
Router(config-router)#network 10.10.10.0 255.255.255.252 area 0
Router(config-router)#network 10.10.10.8 255.255.255.252 area 0
Router(config-router)#network 192.168.3.0 255.2
00:22:16: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.8.1 on Serial0/2/1 from LOADING
to FULL, Loading Done

                                              ^
% Invalid input detected at '^' marker.

Router(config-router)#network 192.168.3.0 255.255.255.0 area 0
Router(config-router)#network 192.168.4.0 255.255.255.0 area 0
Router(config-router)#network 192.168.5.0 255.255.255.0 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
00:24:12: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.2.1 on Serial0/2/0 from LOADING
to FULL, Loading Done
```

(3rd Floor)

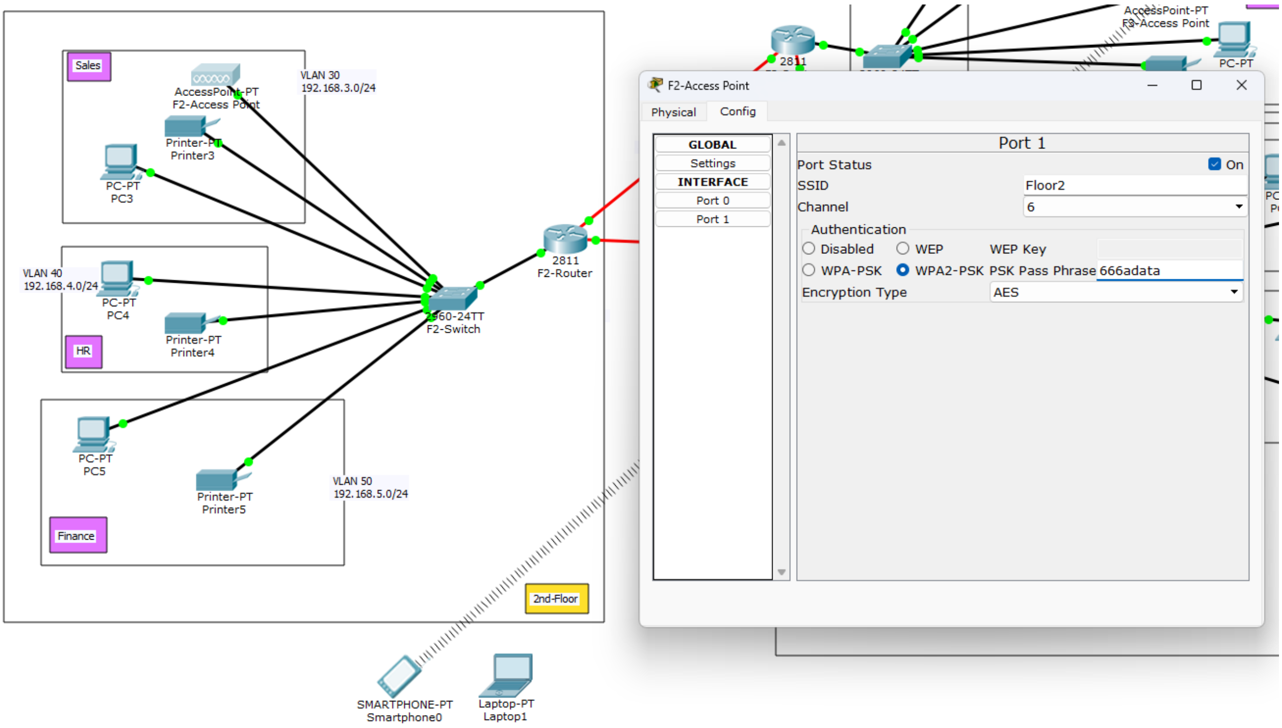
```
Router(config)#router ospf 10
Router(config-router)#network 10.10.10.0 255.255.255.252 area 0
Router(config-router)#network 10.10.10.4 255.255.255.252 area 0
00:24:12: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.5.1 on Serial0/2/1 from LOADING
to FULL, Loading Done

Router(config-router)#network 10.10.10.4 255.255
00:24:25: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.8.1 on Serial0/2/0 from LOADING
to FULL, Loading Done

                                              ^
% Invalid input detected at '^' marker.

Router(config-router)#network 10.10.10.4 255.255.255.252 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#network 192.168.1.0 255.255.255.0 area 0
Router(config-router)#network 192.168.2.0 255.255.255.0 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

Command Prompt

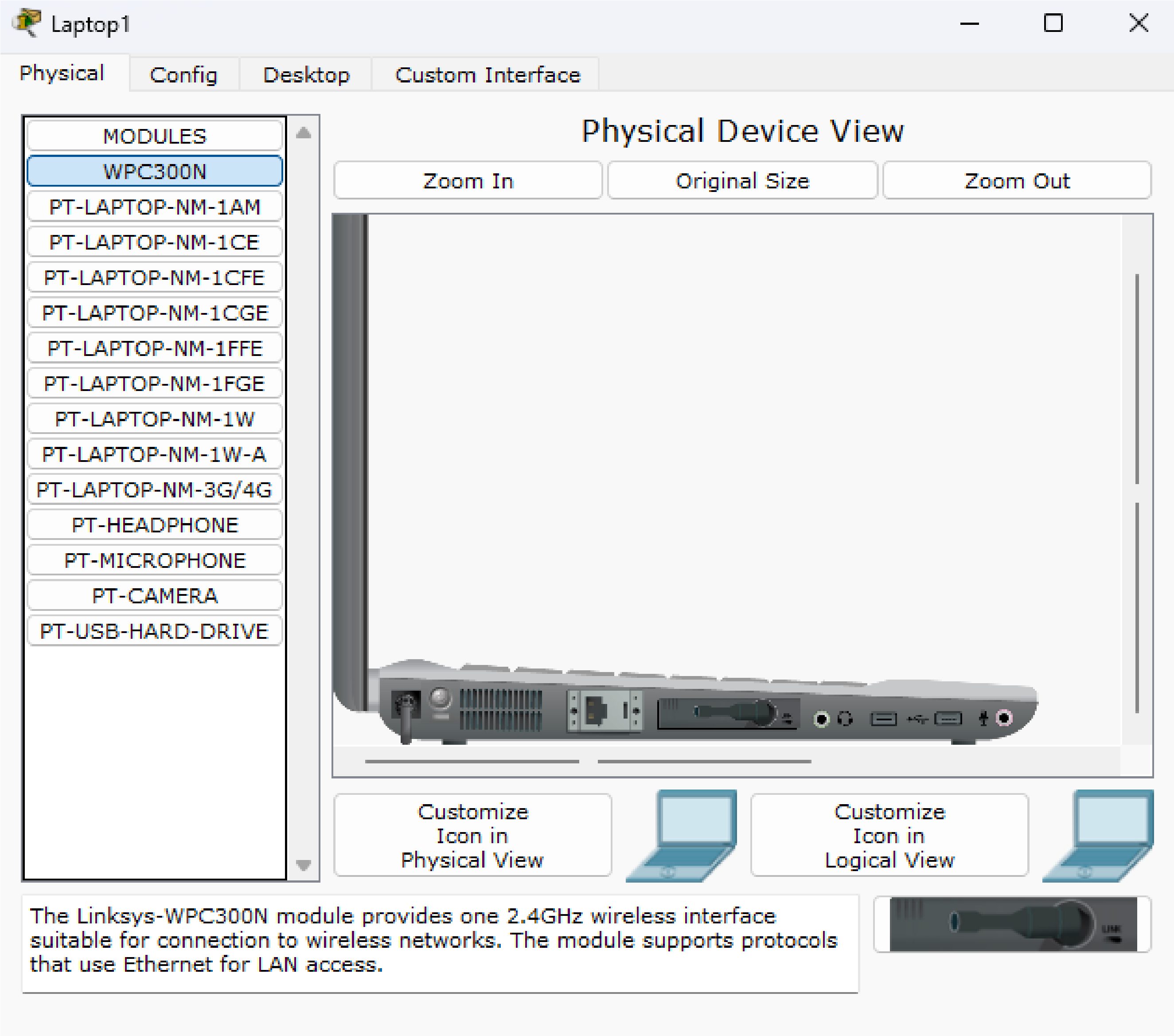
```
PC>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=0ms TTL=127
Reply from 192.168.2.2: bytes=32 time=0ms TTL=127
Reply from 192.168.2.2: bytes=32 time=9ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 3ms

PC>ping 192.168.6.2

Pinging 192.168.6.2 with 32 bytes of data:

Reply from 192.168.6.2: bytes=32 time=5ms TTL=126
Reply from 192.168.6.2: bytes=32 time=8ms TTL=126
Reply from 192.168.6.2: bytes=32 time=1ms TTL=126
Reply from 192.168.6.2: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.6.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 8ms, Average = 4ms

PC>
```

As a result, connection between devices from different floor have been secure.

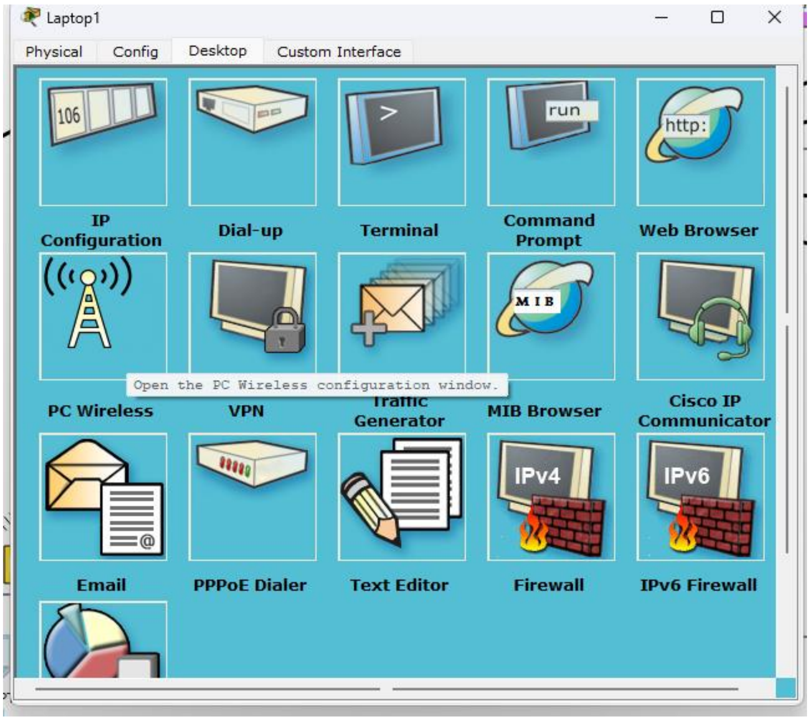
- **Wi-Fi Connections to Laptops & Phones:**



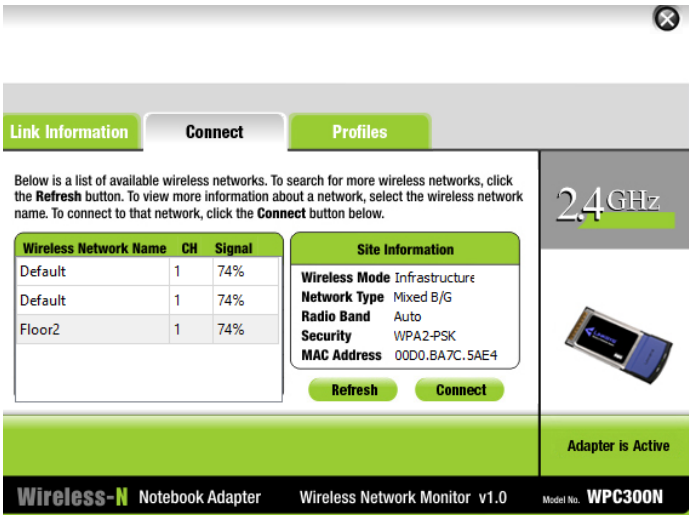
SSID and password for wi-fi being given.

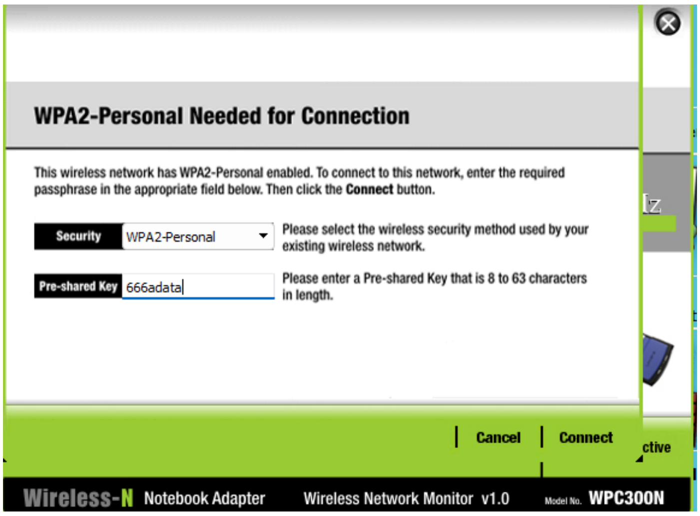


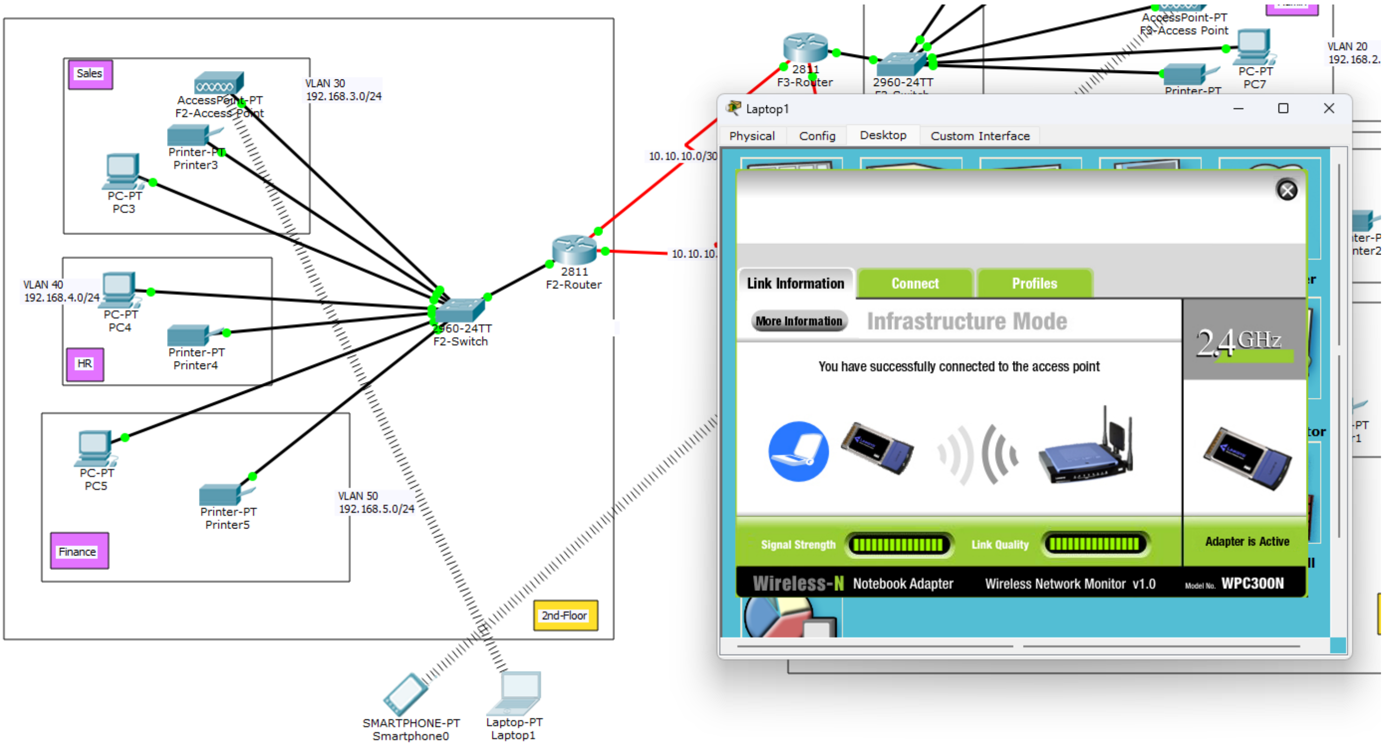
Laptop must be turned off and then WPC300In extension should be put on and then turn on the laptop.



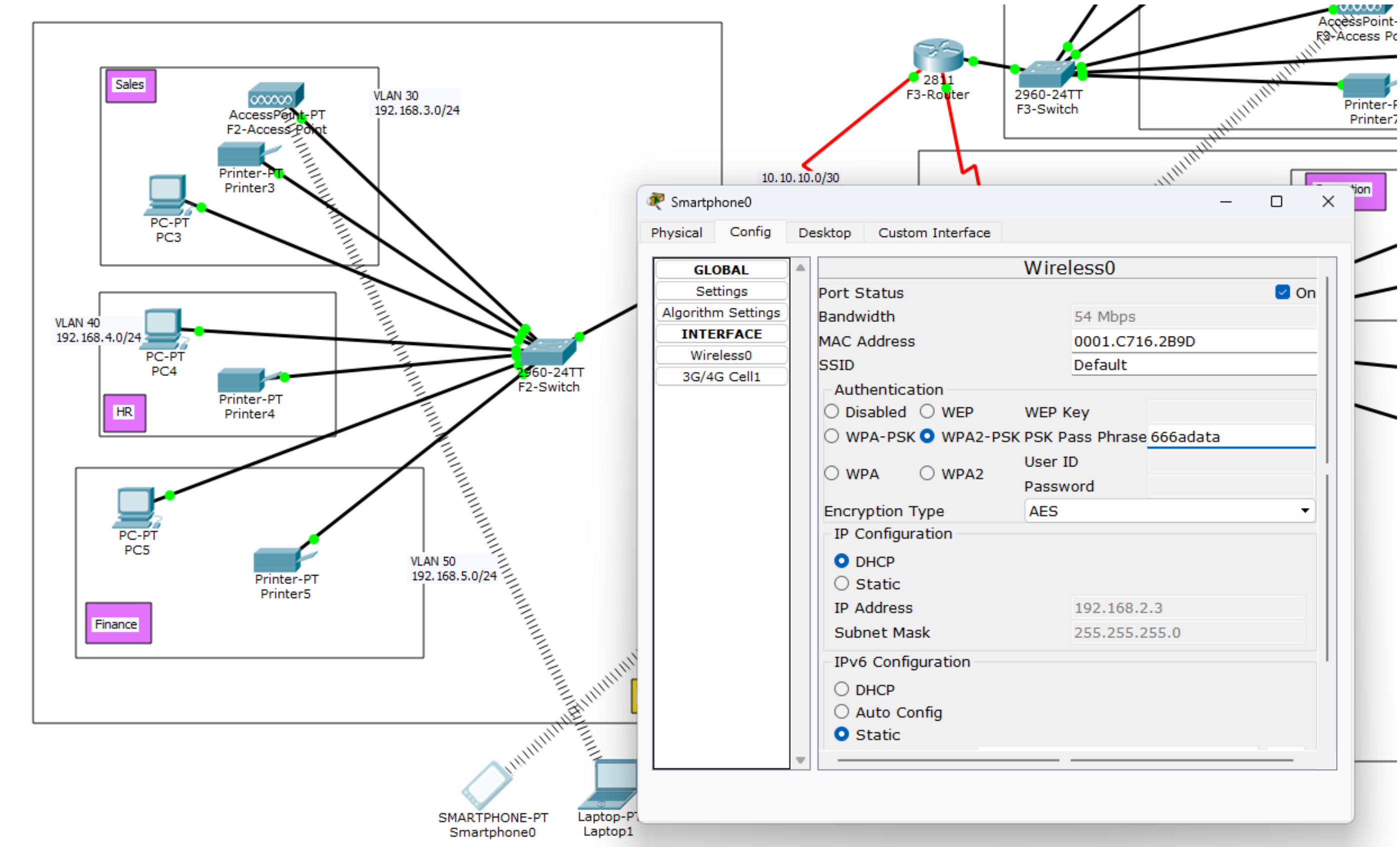
In laptops pc configuration, go to PC wireless and then give refresh and connect to desired wi-fi through password.

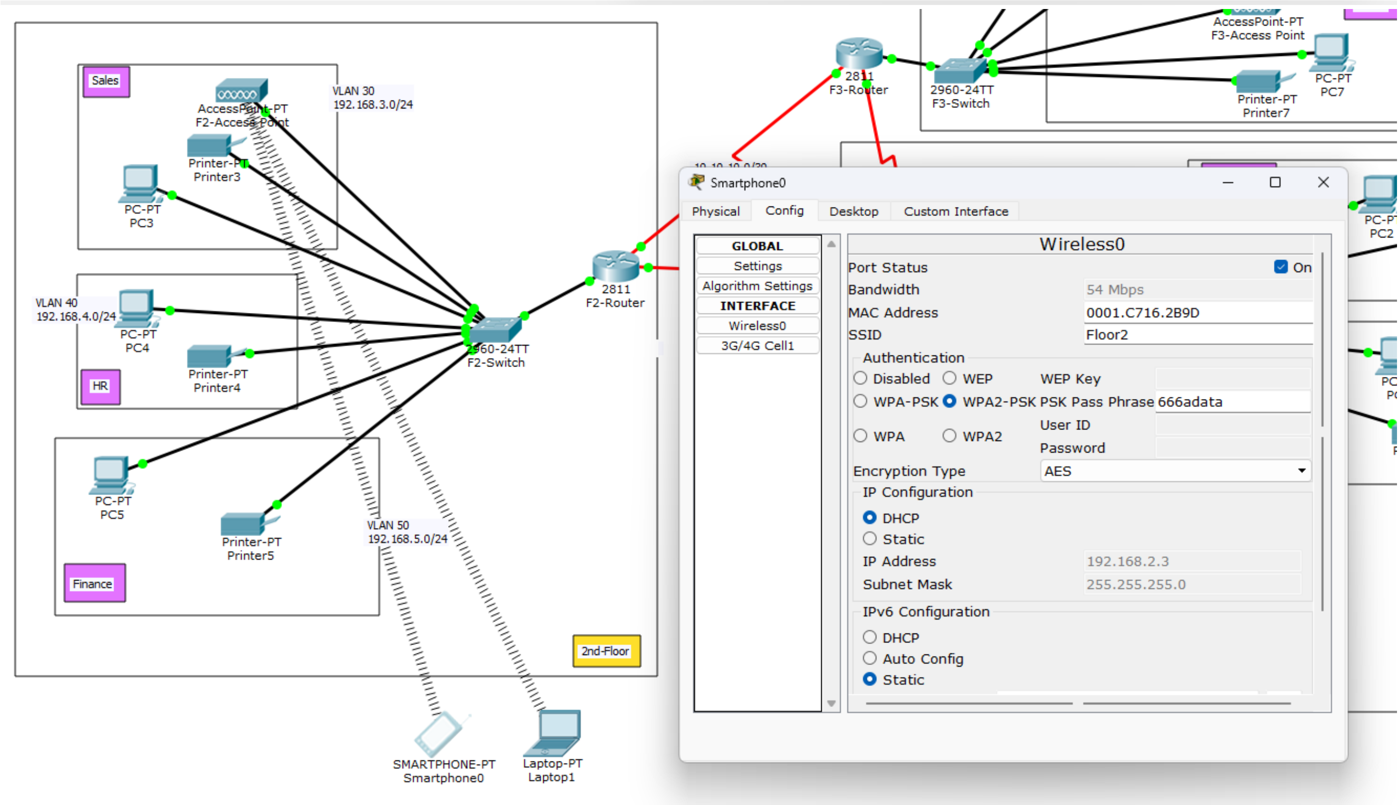






As a result, Laptop is now successfully connected to the wi-fi.



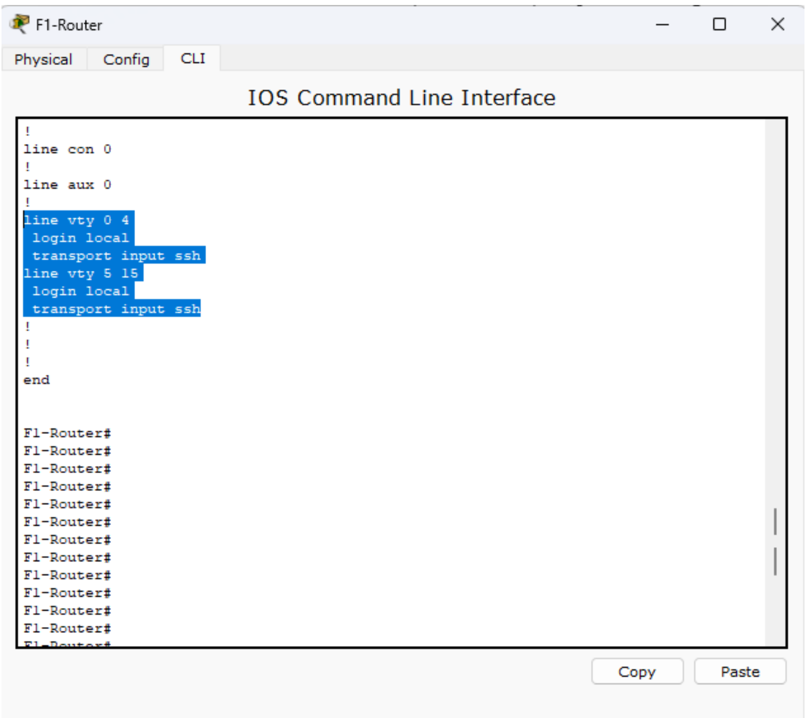


Similar case goes for smartphone.

- **SSH & Port Security**:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname F3-Router
F3-Router(config)#ip domain-name gtech
F3-Router(config)#username gtech password getech
F3-Router(config)#crypto key generate rsa
The name for the keys will be: F3-Router.gtech
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

F3-Router(config)#line vty 0 15
*Mar 1 0:4:48.314:  %SSH-5-ENABLED: SSH 1.99 has been enabled
F3-Router(config-line)#login local
F3-Router(config-line)#transport input ssh
F3-Router(config-line)#do wr
Building configuration...
[OK]
F3-Router(config-line)#ex
% Ambiguous command: "ex"
F3-Router(config-line)#exit
F3-Router(config)#
```

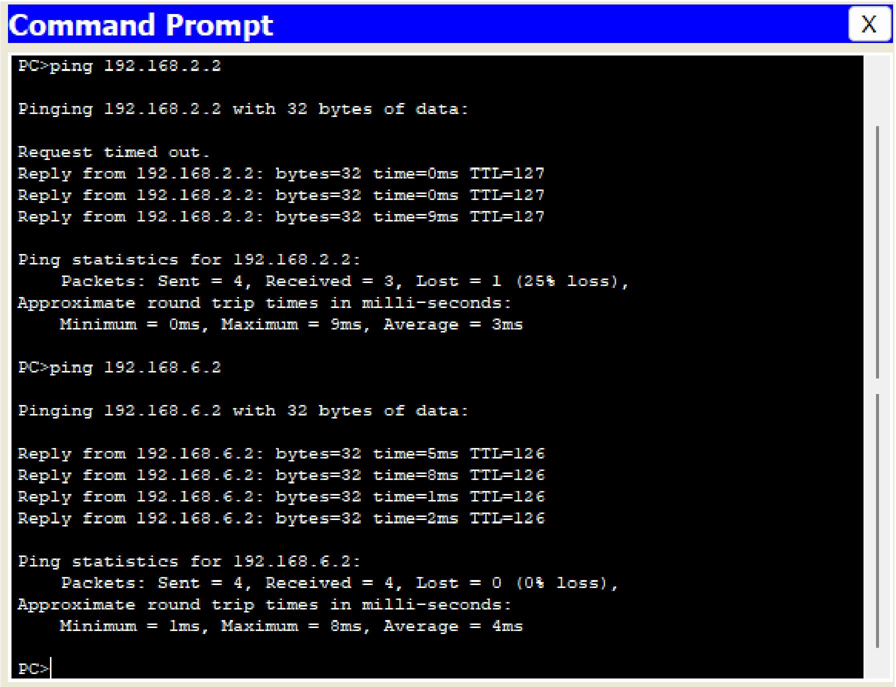
```
Switch(config)#int fa0/2
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#switchport port-security violation ?
  protect   Security violation protect mode
  restrict  Security violation restrict mode
  shutdown  Security violation shutdown mode
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#do wr
Building configuration...
[OK]
```



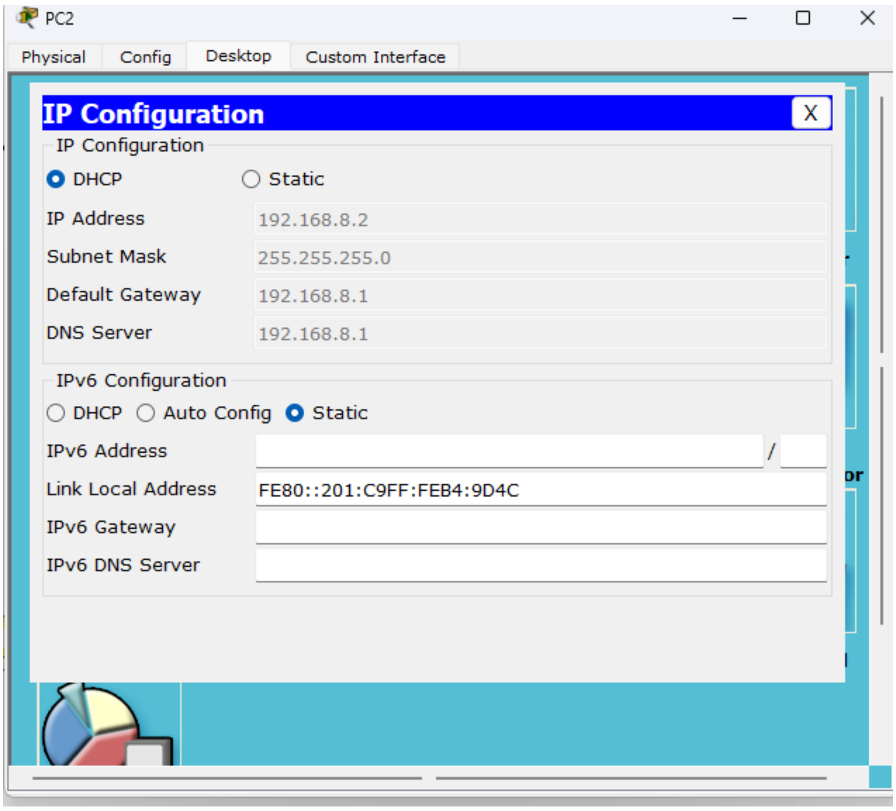
4. Simulation & Results

Tests Conducted

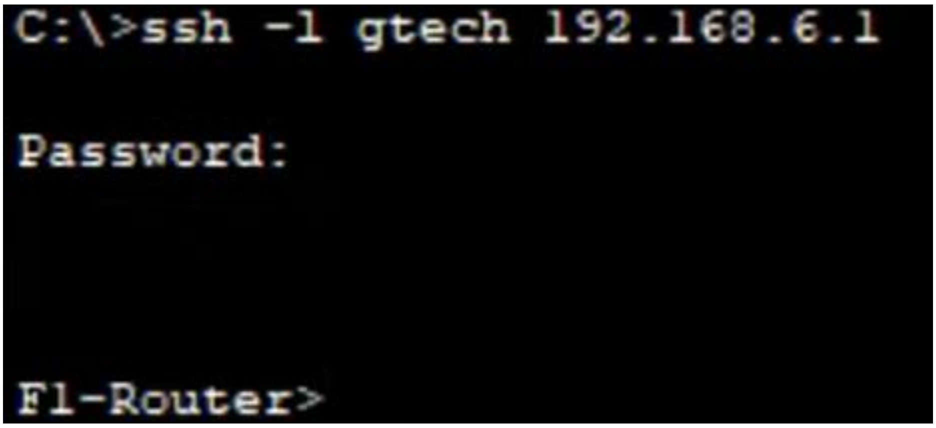
1. **Inter-VLAN Communication**:
    - PCs in **Reception (VLAN 80)** could ping **Finance (VLAN 50)** via OSPF.



2. **DHCP Verification**:
    - All devices received correct IPs from their VLAN’s DHCP pool.

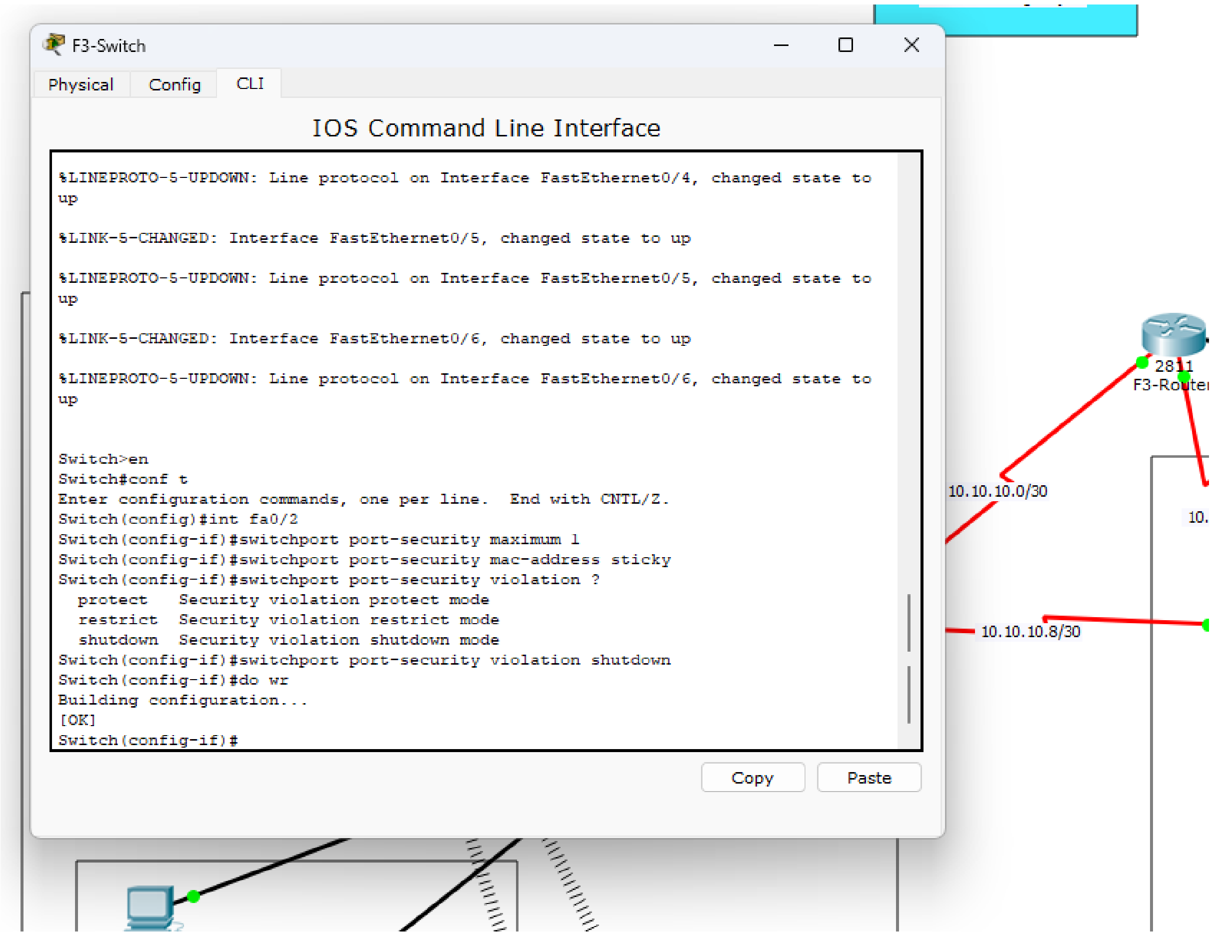


3. **SSH Access**:
    - **Test-PC** in IT successfully logged into routers via SSH.

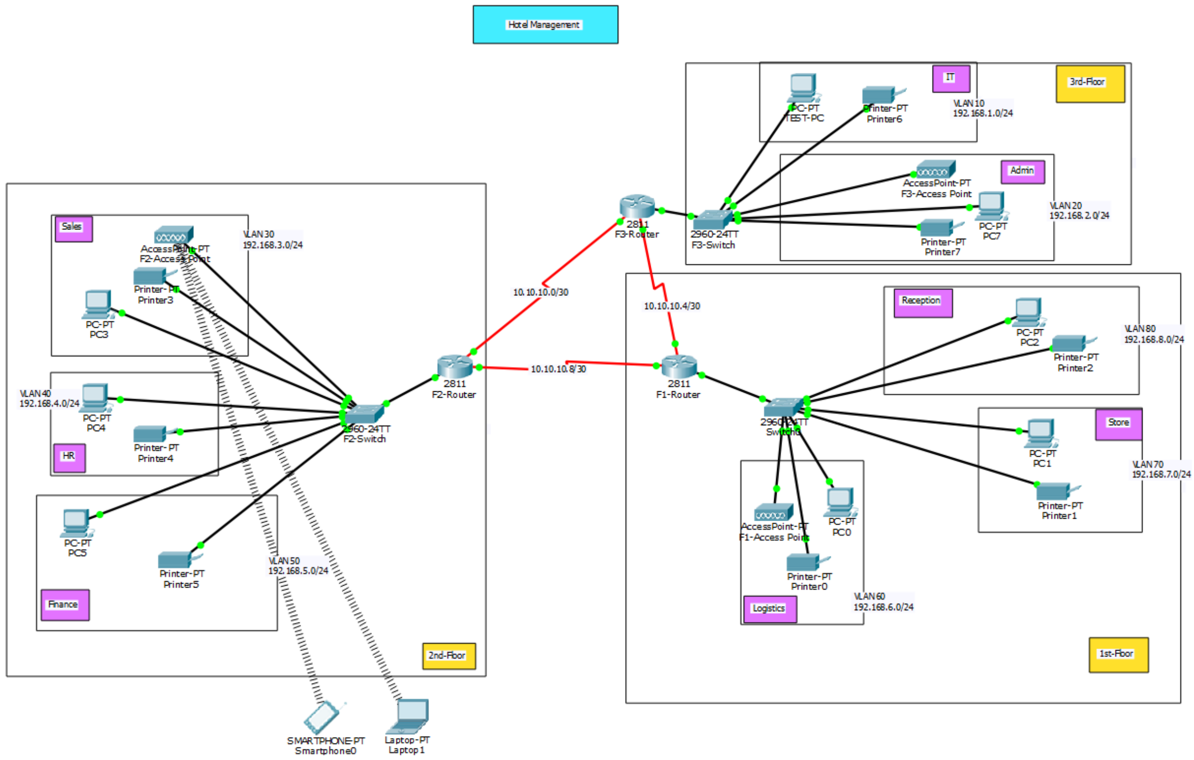


4. **Port Security**:

   - Unauthorized devices on **IT-Switch Fa0/2** triggered port shutdown.



Full Setup:



5. Conclusion

This simulation project successfully demonstrates a realistic and secure network design for a hotel management system. It reflects industry practices in segmenting departmental traffic, ensuring secure communication, and incorporating both wired and wireless devices. The project provided hands-on experience with VLANs, routing, and logical topology design using Cisco Packet Tracer.

The project successfully implemented a **secure, VLAN-segmented hotel network** with:

- **Dynamic routing (OSPF)** for inter-floor connectivity.
- **Controlled access** via port security and SSH.
- **Scalability** for future expansions (e.g., IoT devices).